



National  
Qualifications  
2024

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**2024 Engineering Science**

**National 5**

**Question Paper Finalised Marking Instructions**

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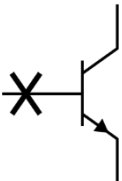
## General marking principles for National 5 Engineering Science

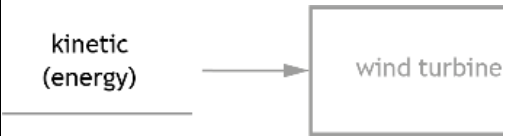
Always apply these general principles. Use them in conjunction with the detailed marking instructions, which identify the key features required in candidates' responses.

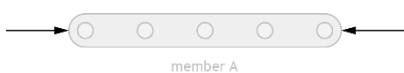
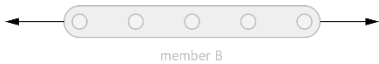
- (a) Always use positive marking. This means candidates accumulate marks for the demonstration of relevant skills, knowledge and understanding; marks are not deducted for errors or omissions.
- (b) Where a candidate makes an error at an early stage in a multi-stage calculation, credit should normally be given for correct follow-on working in subsequent stages, unless the error significantly reduces the complexity of the remaining stages. The same principle should be applied in questions which require several stages of nonmathematical reasoning.
- (c) All units of measurement will be presented in a consistent way, using negative indices where required (eg  $\text{ms}^{-1}$ ). Candidates may respond using this format, or solidus format (m/s) or words (metres per second), or any combination of these (eg metres/second).

## Marking instructions for each question

### Section 1

| Question |     |  | Expected response                                                                                                                                                         | Max mark | Additional guidance                                                                                                            |
|----------|-----|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------------------------------------------------------------------------------------------------------------------------------|
| 1.       | (a) |  | NOT (gate)/Inverter                                                                                                                                                       | 1        | Do not accept symbol.                                                                                                          |
|          | (b) |  |                                                                                          | 1        | Inputs and output must be shown.<br>Accept more than two inputs.                                                               |
| 2.       | (a) |  | (NPN) transistor                                                                                                                                                          | 1        |                                                                                                                                |
|          | (b) |  |                                                                                          | 1        | Accept X close to the base connection or anywhere on the horizontal line.                                                      |
| 3.       | (a) |  | <p>Connection of the main air/exhaust is on the incorrect port.</p> <p>(Pilot air) line type to the cylinder should be solid.</p> <p>The piston should be outstroked.</p> | 2        | <p>1 mark for each error described.</p> <p>1 mark maximum for main air/exhaust error.</p> <p>Accept amendments to diagram.</p> |
|          | (b) |  | Mechanical (engineer)                                                                                                                                                     | 1        |                                                                                                                                |

| Question |     |      | Expected response                                                                                                                                                                                                                                                           | Max mark | Additional guidance                                                                                                                                                                                             |
|----------|-----|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4.       | (a) |      | $\text{velocity ratio} = \frac{\text{speed of input}}{\text{speed of output}}$ $\text{velocity ratio} = \frac{120}{820}$ $\text{velocity ratio} = \frac{6}{41}$ <p><b>VR = 6 : 41 (2 sf)</b></p>                                                                            | 2        | <p>1 mark for substitution.</p> <p>1 mark for correct answer from given working.</p> <p>Ignore any unit.</p> <p>Accept 1 : 6.8<br/>Accept 0.15 : 1<br/>Accept 0.146 (3 sf)</p> <p>Ratio must be simplified.</p> |
|          | (b) |      | <p>No effect.</p> <p>The velocity of gear C will not change.</p>                                                                                                                                                                                                            | 1        |                                                                                                                                                                                                                 |
| 5.       | (a) | (i)  | Civil (engineer)                                                                                                                                                                                                                                                            | 1        |                                                                                                                                                                                                                 |
|          |     | (ii) | Environmental (engineer)                                                                                                                                                                                                                                                    | 1        |                                                                                                                                                                                                                 |
|          | (b) |      |  <p>The diagram shows the text 'kinetic (energy)' on the left, followed by a horizontal arrow pointing to a rectangular box on the right. Inside the box is the text 'wind turbine'.</p> | 1        | Do not accept 'wind'.                                                                                                                                                                                           |
|          | (c) |      | Closed loop (control)                                                                                                                                                                                                                                                       | 1        | Do not accept 'closed' on its own.                                                                                                                                                                              |

| Question |     |      | Expected response                                                                                    | Max mark | Additional guidance                                                                                           |
|----------|-----|------|------------------------------------------------------------------------------------------------------|----------|---------------------------------------------------------------------------------------------------------------|
| 6.       | (a) | (i)  |  <p>member A</p>    | 1        | Accept pairs of forces drawn vertically opposite and facing.                                                  |
|          |     | (ii) |  <p>member B</p>    | 1        | Accept pairs of forces drawn vertically opposite and opposing.<br>Do not accept FTE.                          |
|          | (b) |      | $\sigma = \frac{F}{A}$ $\sigma = \frac{1152}{36}$ $\sigma = 32 \text{ Nmm}^{-2} \text{ (2 sf)}$      | 2        | 1 mark for substitution.<br><br>1 mark for correct answer from given working with unit.                       |
| 7.       | (a) |      | <p>Regular maintenance.</p> <p>Avoid overloading/revving.</p> <p>Lubricate/oil/grease the parts.</p> | 1        | Descriptive answer.<br><br>Do not accept single words.<br><br>Accept use of bearings/use a slippery material. |
|          | (b) |      | Heat/sound (energy)                                                                                  | 1        |                                                                                                               |

| Question |     |      | Expected response                                                                                                                                                                                                                                                                                                                                   | Max mark | Additional guidance                                                                                                                                                                           |
|----------|-----|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 8.       | (a) | (i)  | $R_t = \frac{R_1 \times R_2}{R_1 + R_2}$ $R_t = \frac{390 \times 270}{390 + 270}$ $R_t = 159.5455$ $R_t = 160 \, \Omega \text{ (2 sf)}$ <p>Alternative method</p> $\frac{1}{R_t} = \frac{1}{R_1} + \frac{1}{R_2}$ $\frac{1}{R_t} = \frac{1}{390} + \frac{1}{270}$ $R_t = \frac{1}{0.0062678}$ $R_t = 159.5455$ $R_t = 160 \, \Omega \text{ (2 sf)}$ | 2        | <p>1 mark for substitution.</p> <p>1 mark for correct answer from given working with unit.</p> <p>1 mark for substitution.</p> <p>1 mark for correct answer from given working with unit.</p> |
|          |     | (ii) | $R_t = 780 + 1200 + 160$ $R_t = 2140$ $R_t = 2100 \, \Omega \text{ (2 sf)}$                                                                                                                                                                                                                                                                         | 2        | <p>1 mark for substitution.<br/>Apply FTE from (a)(i).</p> <p>1 mark for correct answer from given working with unit.</p>                                                                     |
|          | (b) |      | $V = I R$ $1.2 = I \times 390$ $I = \frac{1.2}{390}$ $I = 0.00308$ $I = 0.0031 \text{ A (2 sf)}$                                                                                                                                                                                                                                                    | 3        | <p>1 mark for substitution.</p> <p>1 mark for transposition.</p> <p>1 mark for correct answer from given working with unit.</p>                                                               |

| Question |     |      | Expected response                                                                                                                                                                                                                                                                                                                          | Max mark | Additional guidance                                                           |
|----------|-----|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-------------------------------------------------------------------------------|
| 8.       | (c) | (i)  | Voltage $V_1$ - decrease                                                                                                                                                                                                                                                                                                                   | 1        |                                                                               |
|          |     | (ii) | Current $A_1$ - decrease                                                                                                                                                                                                                                                                                                                   | 1        | Apply FTE from part (c)(i).                                                   |
|          | (d) |      | <p>Components are not destroyed<br/>...making it cheaper to test.</p> <p>A range of components are readily available<br/>...making it quicker to change/test.</p> <p>Software can highlight issues in the circuit<br/>...making it easier to correct faults.</p> <p>It is safer to the user<br/>...because fault/failure is simulated.</p> | 2        | <p>Descriptive answer.</p> <p>1 mark for cause.</p> <p>1 mark for effect.</p> |

| Question |     | Expected response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Max mark | Additional guidance                                                                                                                                                                                                                                                                                                                                                                                                       |
|----------|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9.       | (a) | $120 \times 64 = \text{input speed} \times 16$<br><br>$\text{input speed} = \frac{7680}{16}$<br><br>$\text{input speed} = 480 \text{ (revs min}^{-1}\text{)}$<br><br>$480 \times 48 = \text{input speed} \times 16$<br><br>$\text{input speed} = \frac{23040}{16}$<br><br>$\text{input speed} = 1440$<br><br><b>1400 revs min<sup>-1</sup> (2 sf)</b><br><br><b>Alternative method</b><br><br>$\frac{\text{speed of output}}{\text{speed of input}} = \frac{A}{B} \times \frac{C}{D}$<br><br>$\frac{120}{\text{input speed}} = \frac{16}{48} \times \frac{16}{64}$<br><br>$\frac{120}{\text{input speed}} = \frac{1}{12}$<br><br>$\text{input speed} = 120 \times \frac{12}{1}$<br><br>$\text{input speed} = 1440$<br><br><b>1400 revs min<sup>-1</sup> (2 sf)</b> | 4        | 1 mark for substitution.<br><br><br>1 mark for correct answer from given working (unit not required).<br><br>1 mark for substitution.<br>Apply FTE.<br><br><br>1 mark for correct answer from given working with unit.<br><br>Do not accept 'RPM'.<br><br><br>1 mark for first ratio.<br>1 mark for second ratio.<br><br><br>1 mark for transposition.<br><br><br>1 mark for correct answer from given working with unit. |

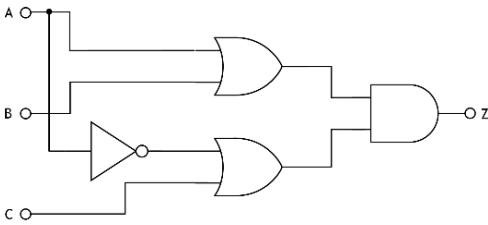


| Question |     |  | Expected response                                                                                                                                                                                                                                                           | Max mark | Additional guidance                                                                                                                       |
|----------|-----|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------|
| 9.       | (b) |  | $P = VI$<br>$250 = 36 \times I$<br>$I = \frac{250}{36}$<br>$I = 6.944444444$<br><b><math>I = 6.9 \text{ A (2 sf)}</math></b>                                                                                                                                                | 3        | 1 mark for substitution.<br>1 mark for transposition.<br><br>1 mark for correct answer from given working with unit.                      |
|          | (c) |  | Lubricate/oil the gears<br>...to reduce friction/energy loss.                                                                                                                                                                                                               | 2        | Descriptive answer.<br><br>1 mark for cause with reference to the gear train.<br><br>1 mark for effect.                                   |
|          | (d) |  | No CO <sub>2</sub> emissions/pollution (when cycling).<br><br>Fewer car journeys so less noise/pollution.                                                                                                                                                                   | 1        | Descriptive answer relating to use with a positive environmental aspect.                                                                  |
|          | (e) |  | Easier/less effort, so encourages more people to cycle.<br><br>Improves fitness/lifestyle/wellbeing.<br><br>Journey times reduced.<br><br>Reduces the number of cars on the road, making it safer to travel.<br><br>Reduction in noise for residents living next to a road. | 1        | Descriptive answer relating to use with an improvement to life aspect.                                                                    |
|          | (f) |  | Expensive to maintain/repair/hire the electric bike.<br><br>Maintenance of charging stations is expensive.<br><br>Expensive to charge.<br><br>Reduction in car sales.                                                                                                       | 1        | Descriptive answer relating to use with a negative monetary aspect.<br><br>Responses can refer to renting versus owning an electric bike. |

| Question |     |  | Expected response                                                                                                                                                                                                                                                                                                     | Max mark | Additional guidance                                                                                                                                                                                                                                                                                                                               |
|----------|-----|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10.      | (a) |  | 800 $\Omega$                                                                                                                                                                                                                                                                                                          | 1        | 1 mark for correct answer with unit.                                                                                                                                                                                                                                                                                                              |
|          | (b) |  | Voltage divider.                                                                                                                                                                                                                                                                                                      | 1        | Accept potential divider.                                                                                                                                                                                                                                                                                                                         |
|          | (c) |  | $V_{\text{therm}} = 5 - 1.4$<br>$V_{\text{therm}} = 3.6 \text{ V}$<br>$\frac{R_1}{R_2} = \frac{V_1}{V_2}$<br>$\frac{R_1}{4.8\text{k}} = \frac{1.4\text{V}}{3.6\text{V}}$<br>$R_1 = \frac{1.4\text{V}}{3.6\text{V}} \times 4.8\text{k}$<br>$R_1 = 1.866666667$<br><b><math>R_1 = 1.9 \text{ k}\Omega</math> (2 sf)</b> | 4        | 1 mark $V_{\text{therm}}$ unit not required.<br><br>1 mark for substitution.<br>Apply FTE for $V_{\text{therm}}$ .<br>If $V_{\text{therm}}$ not calculated, accept 5V for substitution.<br><br>1 mark for transposition.<br><br>1 mark for correct answer from given working with unit.<br><br>Accept alternative method.                         |
|          | (d) |  | <i>As the water temperature increases...</i><br><br>The resistance of the thermistor will decrease.<br><br>This will cause $V_{\text{in}}$ to increase.<br><br>The transistor will then switch on/ the relay will activate.<br><br>The blue LED will turn off<br><br>and the red LED will switch on.                  | 5        | Descriptive answer.<br>Apply FTE from previous description.<br><br>1 mark thermistor resistance decrease.<br><br>1 mark $V_{\text{in}}$ increase.<br><br>1 mark transistor/relay switching on.<br><br>1 mark blue LED off.<br><br>1 mark red LED on.<br><br>Award a maximum of 1 mark for both LEDs (red & blue) switch on/ both LEDs switch off. |

| Question |     |  | Expected response                                                                                                                                                                                                                     | Max mark | Additional guidance                                                                                                                                                                                                                                                                                                    |
|----------|-----|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10.      | (e) |  | <p><u>Ammonia fuelled engines</u> are new and untried</p> <p>... so they may be unreliable.</p> <p><u>Nuclear battery</u> charge lasts up to 50 years</p> <p>... so portable product will never need recharged in their lifetime.</p> | 2        | <p>Descriptive answer.</p> <p>1 mark for cause.</p> <p>1 mark for effect.</p> <p>If an established or developing technology is named, such as AI or self-driving cars, 1 mark maximum for cause <b>and</b> effect.</p> <p>No mark for only stating an emerging technology.</p>                                         |
| 11.      | (a) |  |                                                                                                                                                                                                                                       | 4        | <p>1 mark - set temperature<br/>Accept user input.</p> <p>1 mark - temperature sensor.<br/>Accept heat sensor/thermostat.<br/>Do not accept thermometer/thermistor.</p> <p>1 mark - feedback loop with both arrows starting after heating element.</p> <p>1 mark - heating element/heater.<br/>Do not accept heat.</p> |
|          |     |  |                                                                                                                                                                                                                                       |          |                                                                                                                                                                                                                                                                                                                        |

| Question |     | Expected response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Max mark | Additional guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 11.      | (b) | <pre> graph TD     Start([start]) --&gt; Pin2{is pin 2 on?}     Pin2 -- N --&gt; Pin7On[/pin 7 on/]     Pin2 -- Y --&gt; Pin1{is pin 1 on?}     Pin1 -- N --&gt; Pin2     Pin1 -- Y --&gt; Pin6On[/pin 6 on/]     Pin6On --&gt; Wait35[wait 35 s]     Wait35 --&gt; Pin6Off[/pin 6 off/]     Pin6Off --&gt; Pin5On[/pin 5 on/]     Pin5On --&gt; Wait300[wait 300 s]     Wait300 --&gt; Pin5Off[/pin 5 off/]     Pin5Off --&gt; Pin4On[/pin 4 on/]     Pin4On --&gt; Wait03[wait 0.3 s]     Wait03 --&gt; Pin4Off[/pin 4 off/]     Pin4Off --&gt; Wait03_2[wait 0.3 s]     Wait03_2 --&gt; Loop{loop 6 times?}     Loop -- N --&gt; Pin2     Loop -- Y --&gt; Pin7Off[/pin 7 off/]     Pin7Off --&gt; Start </pre> | 11       | <p>Pin 7 on and off in correct position - 1 mark.</p> <p>Pin 1 on? with y/n - 1 mark.</p> <p>Pin 1 loop back with arrow before decision - 1 mark.</p> <p>Pin 6 on and off in correct position - 1 mark.</p> <p>Pin 5 on and off in correct position - 1 mark.</p> <p>Pin 4 on and off in correct position - 1 mark.</p> <p>All four delay times with correct units - 1 mark.</p> <p>Accept 5 minutes rather than 300 s.</p> <p>Loop x 6? with y/n - 1 mark.</p> <p>Feedback loop and arrow to before pin 4 on - 1 mark.</p> <p>Continuous loop and arrow to start of program - 1 mark.</p> <p>All symbols correct - 1 mark.</p> <p>Ignore any additional steps including their symbols.</p> |

| Question |     |  | Expected response                                                                                                                                                                                                                             | Max mark | Additional guidance                                                                                                                                                                                                                                                                                                              |
|----------|-----|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12.      | (a) |  | <p>Using solar panels, generate energy/<br/>do not use fossil fuels</p> <p>...so no greenhouse gases will be released.</p>                                                                                                                    | 2        | <p>Descriptive response.</p> <p>1 mark for cause (generate energy from sun/reduced use of fossil fuels).</p> <p>1 mark for effect (no release of greenhouse gasses).</p> <p>Do not accept pollution.</p> <p>Accept carbon emissions.</p> <p>Gases/emission must be specific to greenhouse or CO<sub>2</sub>/dN<sub>2</sub>O.</p> |
|          | (b) |  | <p><math>7.5 \times 60 \times 60 = 27000 \text{ (s)}</math></p> <p><math>E_e = VIt</math></p> <p><math>E_e = 24 \times 0.42 \times 27000</math></p> <p><math>E_e = 272160</math></p> <p><b><math>E_e = 270000 \text{ J (2 sf)}</math></b></p> | 3        | <p>1 mark for time, units not required.</p> <p>1 mark for substitution.</p> <p>Apply FTE from time calculation. If time in seconds not calculated, accept 7.5 hrs for substitution.</p> <p>1 mark for correct answer from given working with unit.</p>                                                                           |
|          | (c) |  |                                                                                                                                                            | 4        | <p>1 mark for input A connected to a NOT gate.</p> <p>1 mark for both input A and input B directly connected to an OR gate.</p> <p>1 mark for both input (NOT) A and input C connected to an OR gate.</p> <p>1 mark for AND gate output directly wired to Z with input connections.</p>                                          |

| Question |     |       | Expected response                                                                                                                                                                                       | Max mark | Additional guidance                                                                                                                                             |
|----------|-----|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 13.      | (a) | (i)   | Design the propeller.<br><br>Calculate the output speed of the propeller.<br><br>Test the drive system of the propeller.<br><br>Simulate the water flow over the wing.                                  | 1        | Descriptive answer.<br><br>Activity and mechanical aspect.                                                                                                      |
|          |     | (ii)  | Calculate the forces acting on the mast/board/wing.<br><br>Selecting materials for the board.<br><br>Design the shape of the wing/mast.<br><br>Simulate the forces acting on the mast/board/wing.       | 1        | Descriptive answer.<br><br>Activity and structural aspect.                                                                                                      |
|          |     | (iii) | Design the controller's wireless/control circuit.<br><br>Simulate the control circuit.<br><br>Write the control program for the wireless controller.<br><br>Select sensors for the wireless controller. | 1        | Descriptive answer.<br><br>Activity and electronics aspect.<br><br>Do not accept 'design the controller' on its own.                                            |
|          | (b) |       | $E_k = \frac{1}{2} mv^2$<br><br>$13000 = \frac{1}{2} \times 115 \times v^2$<br><br>$v = \sqrt{\frac{2 \times 13000}{115}}$<br><br>$v = 15.0362$<br><br>$v = 15 \text{ ms}^{-1} \text{ (2 sf)}$          | 3        | 1 mark for substitution.<br><br>1 mark for transposition.<br><br>1 mark for correct answer from given working with unit.                                        |
|          | (c) |       | C<br><br>It is corrosion resistant (against the water) and high strength (resisting the range of forces acting on it).                                                                                  | 2        | 1 mark for selecting material C.<br><br>1 mark for justification that includes reference to corrosion resistant and high strength.<br><br>Do not accept strong. |

| Question |     |      | Expected response                                                                                                                                                                                                 | Max mark | Additional guidance                                                                                                             |
|----------|-----|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------|
| 14.      | (a) | (i)  | $\Sigma_{\text{moments}} = 0$<br>$\Sigma CWM = \Sigma ACWM$<br>$(46k \times 4.5) = (3.9k \times 10) + (R_1 \times 11.2)$<br>$207 = 39 + 11.2R_1$<br>$R_1 = \frac{207 - 39}{11.2}$<br>$R_1 = 15 \text{ kN (2 sf)}$ | 3        | <p>1 mark for substitution.</p> <p>1 mark for transposition.</p> <p>1 mark for correct answer from given working with unit.</p> |
|          |     | (ii) | $\Sigma F_{\text{vertical}} = 0$<br>$\Sigma F_{\text{up}} = \Sigma F_{\text{down}}$<br>$15k + 3.9k + R_2 = 46k$<br>$R_2 = 46k - 18.9k$<br>$R_2 = 27.1$<br>$R_2 = 27 \text{ kN (2 sf)}$                            | 2        | <p>1 mark for substitution</p> <p>Apply FTE from (a)(i).</p> <p>1 mark for correct answer from given working with unit.</p>     |
|          | (b) |      | $\epsilon = \frac{\Delta l}{l}$<br>$0.00038 = \frac{0.46}{l}$<br>$l = \frac{0.46}{0.00038}$<br>$l = 1210.526316$<br>$l = 1200 \text{ mm (2 sf)}$                                                                  | 3        | <p>1 mark for substitution.</p> <p>1 mark for transposition.</p> <p>1 mark for correct answer from given working with unit.</p> |

| Question |     |   | Expected response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Max mark | Additional guidance                                                                                                                                                                                                                                                                                                                                                                                                                                              |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                  |
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| 14.      | (c) |   | <table><tr><td>A</td><td>B</td><td>C</td><td>D</td><td>E</td><td>F</td><td>Z</td></tr><tr><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td></tr><tr><td>0</td><td>0</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td><td>1</td><td>0</td></tr><tr><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td><td>1</td><td>1</td></tr><tr><td>1</td><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td></tr></table> | A        | B                                                                                                                                                                                                                                                                                                                                                                                                                                                                | C | D | E | F | Z | 0 | 0 | 0 | 1 | 1 | 1 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 1 | 0 | 0 | 0 | 1 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 3 | 1 mark for each correct column.<br><br>1 mark for column E = NOT B.<br><br>1 mark for column F = D OR E.<br><br>1 mark for column Z = F AND C.<br><br>Apply FTE. |
| A        | B   | C | D                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | E        | F                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Z |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                  |
| 0        | 0   | 0 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 1        | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 0 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                  |
| 0        | 0   | 1 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 1        | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 1 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                  |
| 0        | 1   | 0 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 0        | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 0 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                  |
| 0        | 1   | 1 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 0        | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 1 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                  |
| 1        | 0   | 0 | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 1        | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 0 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                  |
| 1        | 0   | 1 | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 1        | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 1 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                  |
| 1        | 1   | 0 | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 0        | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 0 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                  |
| 1        | 1   | 1 | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 0        | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 0 |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                  |
| 15.      | (a) |   | <p>When the air bleed is covered valve 1 is actuated ...</p> <p>Pilot air changes the state of valve 2 which will cause piston B to outstroke.</p> <p>Piston B will actuate valve 3 (sending air to cylinder A) which will cause piston A to outstroke.</p> <p>When valve 3 is actuated a time delay will occur/a time delay will occur before changing the state of valve 2.</p> <p>(Pilot air) changes the state of valve 2 which will cause piston B to instroke.</p> <p>When plunger on valve 3 is unactuated, piston A will instroke.</p>                                                                                                                                                                                                         | 5        | Descriptive response.<br><br>1 mark for changes the state of valve 2 and B outstroking.<br><br>1 mark for actuating valve 3 and outstroking A.<br><br>1 mark for time delay (stated or inferred) after valve 3 actuates/ after A outstrokes/before valve 2 resets.<br><br>1 mark for changes the state of valve 2 and instroking B.<br><br>1 mark for A instroking including condition stated or inferred.<br><br>Accept actuate as change of state for valve 2. |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                  |



| Question |     |  | Expected response                                                                                                                                                                                                             | Max mark | Additional guidance                                                                                                                                                                                                                                                        |
|----------|-----|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15.      | (b) |  | $A = \frac{\pi d^2}{4}$ $A = \frac{\pi 12^2}{4}$ $A = 113.0973355 \text{ (mm}^2\text{)} \quad (36\pi)$ $P = \frac{F}{A}$ $P = \frac{18 \text{ N}}{113.0973355}$ $P = 0.1591549431$ $P = 0.16 \text{ Nmm}^{-2} \text{ (2 sf)}$ | 3        | <p>1 mark for area (unit not required)<br/>Accept <math>36\pi</math>.</p> <p>1 mark for substitution (apply FTE from area).<br/>Accept use of diameter (12 mm) for substitution if area not calculated.</p> <p>1 mark for correct answer from given working with unit.</p> |
|          | (c) |  | <p>Slow the air<br/>...on the 5/2 valve's exhaust port.</p> <p>Add a uni-directional restrictor<br/>...air in the instroking pipe of the double acting cylinder.</p>                                                          | 2        | <p>Descriptive response.</p> <p>1 mark for restricting the air flow.</p> <p>1 mark for exhaust air location.</p> <p>Accept responses shown on the circuit diagram.</p> <p>Do not accept 'UDR' on its own.</p>                                                              |
|          | (d) |  | Reciprocating                                                                                                                                                                                                                 | 1        | Accept reciprocal.                                                                                                                                                                                                                                                         |

[END OF MARKING INSTRUCTIONS]